```
library(ggplot2)

games <- read.csv(file = "vgsales.csv",
                  header = T,
                  stringsAsFactors = F)
names(games)

# Have to see the variables I can use from the dataset^

# Which video game genres have the highest global sales?
install.packages("dplyr")
library(dplyr)


genre_sales <- games %>%
  group_by(Genre) %>%
  summarise(Global_Sales = sum(Global_Sales, na.rm = T))


ggplot(genre_sales, aes(x = reorder(Genre, Global_Sales), y = Global_Sales)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Global Video Game Sales by Genre",
    x = "Genre",
    y = "Global Sales (Millions)"
  )



# Which gaming platforms have the highest global sales?
platform_sales <- games %>%
  group_by(Platform) %>%
  summarise(Global_Sales = sum(Global_Sales, na.rm = TRUE)) %>%
  arrange(desc(Global_Sales)) %>%
  slice_head(n = 10)

ggplot(platform_sales, aes(x = reorder(Platform, Global_Sales), y = Global_Sales)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top 10 Platforms by Global Video Game Sales",
    x = "Platform",
    y = "Global Sales (Millions)"
  )

# How do video game genres and platforms relate to global sales?


platform_genre_sales <- games %>%
  group_by(Platform, Genre) %>%
  summarise(
    Global_Sales = sum(Global_Sales, na.rm = TRUE)
  )

top_platforms <- platform_genre_sales %>%
  group_by(Platform) %>%
  summarise(Total_Sales = sum(Global_Sales)) %>%
  arrange(desc(Total_Sales)) %>%
  slice_head(n = 10)        #Top 10 platforms
```

```
platform_genre_sales <- platform_genre_sales %>%
  filter(Platform == "PS2" | Platform == "X360" | Platform == "PS3" |
           Platform == "Wii" | Platform == "DS" | Platform == "PS" |
           Platform == "GBA" | Platform == "PSP" | Platform == "PS4" |
           Platform == "PC")

ggplot(platform_genre_sales, aes(x = Platform, y = Global_Sales, fill = Genre)) +
  geom_col() +
  labs(
    title = "Global Video Game Sales by Platform and Genre",
    x = "Platform",
    y = "Global Sales (Millions)",
    fill = "Genre"
  ) +
  theme_minimal()

# Do different regions show different preferences for video game genres?

library(tidyverse)

regional_sales <- games %>%
  group_by(Genre) %>%
  summarise(
    North_America = sum(NA_Sales, na.rm = TRUE),
    Europe = sum(EU_Sales, na.rm = TRUE),
    Japan = sum(JP_Sales, na.rm = TRUE),
    Other = sum(Other_Sales, na.rm = TRUE)
  ) %>%
  pivot_longer(
    cols = c(North_America, Europe, Japan, Other),
    names_to = "Region",
    values_to = "Sales"
  )

ggplot(regional_sales, aes(x = Genre, y = Sales, fill = Region)) +
  geom_col(position = "dodge") +
  coord_flip() +
  labs(
    title = "Video Game Sales by Genre and Region",
    x = "Genre",
    y = "Sales (Millions)",
    fill = "Region"
  ) +
  theme_minimal()

ggsave("Sales_by_GR_NL.pdf")


# Main Question: What Factors are associated with higher video game sales?

main_data <- games[games$Year != "N/A", ]
main_data$Year <- as.numeric(main_data$Year)

main_data <- main_data[main_data$Platform == "DS" |
                         main_data$Platform == "PS2" |
                         main_data$Platform == "PS3" |
                         main_data$Platform == "Wii" |
                         main_data$Platform == "X360" |
                         main_data$Platform == "PSP", ]
```

```
ggplot(main_data, aes(x = Year, y = Global_Sales, color = Genre)) +
  geom_point(alpha = 0.5) +
  facet_wrap(~ Platform) +
  labs(
    title = "Global Video Game Sales by Release Year, Genre, and Platform",
    x = "Release Year",
    y = "Global Sales (Millions)",
    color = "Genre"
  ) +
  theme_minimal()
```